Spatial Graph Attention and Curiosity-driven Policy for Antiviral Drug Discovery

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Overview

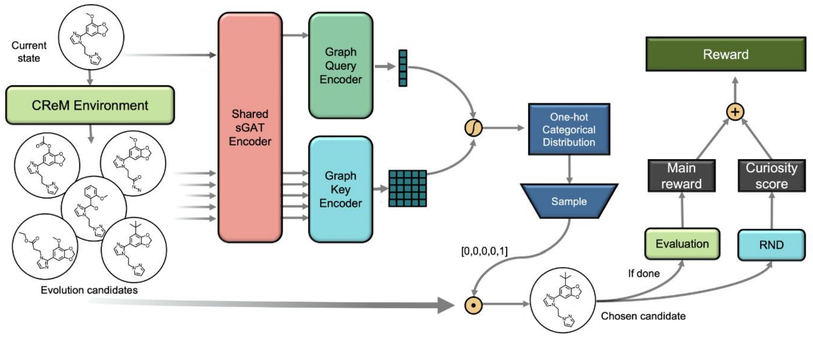
Distilled Graph Attention Policy Network (DGAPN) is a reinforcement learning model for generating novel graph-structured chemical representations that optimize user-defined objectives by efficiently navigating a constrained domain

The framework is examined on the task of generating molecules that are designed to bind, noncovalently, to functional sites of a SARS-CoV-2 protein.

Our proposed method:

- uses state-of-the-art policy gradient reinforcement learning framework
- defines fragment-interchanging-based action space that is chemically reasonable
- embeds both atom and bond attributes as well as spatial structures by leveraging self-attention
- incorporates an efficient surrogate approach that proposes curiosity bonus to enhance exploration
- is parallelized in various fashions to fit specific needs

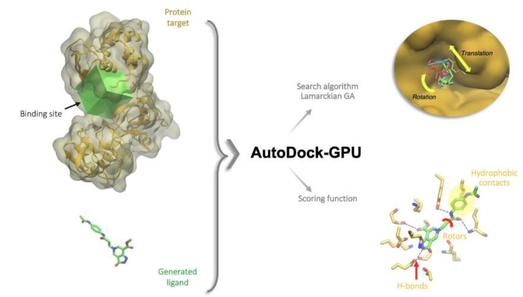
Distilled Graph Attention Policy Network



Overview of DGAPN during a single step of the generating process

The generating process is framed as a Markov decision problem. In a single step, the CReM environment proposes candidates that are reasonable evolutions for the current molecular state; the current state and candidates are encoded into hidden graph representations by a shared Spatial Graph Attention (sGAT) network, then encoded into vector representations by two separate networks; a retrieval system attention mechanism is applied to predict the selection probability of each candidate; a candidate is sampled as the next state and its curiosity score is evaluated by random network distillation [1] as part of the reward; if stopping criteria are met, the main objective is evaluated and added to the reward. After a certain number of steps and rewards are recorded, the network is updated by the proximal policy gradient [2] technique.

Reward from Protein-Ligand Structural Information



The main reward of a generated ligand is its binding affinity estimate against the catalytic site of SARS-CoV-2 protein NSP15, predicted by molecular docking.

Experiments

Single-objective optimization

The negative docking score is used as the main reward r_m and assign it to the final state s_T as the single objective.

Table 1: Primary objective and other summary metrics in evaluations

	Dock Score								
	mean	1st	2nd	3rd	Validity	Diversity	QED	SA	FCD
REINVENT	-5.6	-10.22	-9.76	-9.50	95%	0.88	0.57	7.80	7e-3
JTVAE	-5.6	-8.56	-8.39	-8.39	**100%**	0.86	**0.70**	3.34	2e-1
GCPN	-5.0	-9.19	-7.93	-7.54	**100%**	**0.91**	0.44	5.48	**8e-4**
MolDQN	-7.5	-10.59	-10.42	-10.17	**100%**	0.87	0.17	5.01	3e-3
MARS	-6.2	-10.29	-9.98	-9.88	**100%**	0.88	0.27	3.65	4e-2
DGAPN	**-8.3**	**-12.22**	**-12.11**	**-11.62**	**100%**	0.86	0.14	**3.19**	6e-3

DGAPN managed to generate molecules with improved docking scores compared to state-of-the-art models

Ablation studies

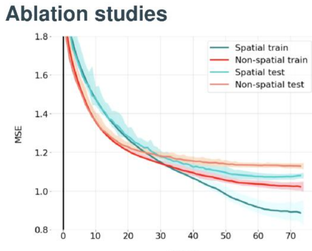


Figure 3: Loss in supervised learning with and without spatial convolution

	Dock Score					
	mean	1st	2nd	Diversity	QED	SA
GAPN	-8.1	-11.37	-11.33	**0.83**	**0.20**	**3.12**
DGAPN	**-8.3**	**-12.22**	**-12.11**	0.86	0.14	3.19
CReM Greedy	-8.8	-10.60	-10.50	**0.85**	**0.28**	**2.99**
DGAPN eval	**-9.3**	**-13.89**	**-13.33**	0.84	0.26	3.44

Table 2: Dock scores and other metrics under different training and evaluation settings

Experiments

Constrained optimization

In lead optimization problems in drug discovery and material science, it can be useful to require the generated molecule to maintain a certain level of similarity to its starting molecule.

$$r_{m'}(s_T) = r_m(s_T) - \lambda \cdot \max\{0, \delta - SIM\{s_0, s_T\}\}$$

The three best performing models in single-objective optimization are compared.

Table 3: Objective improvements and molecule similarities under different constraining coefficients

	MolDQN		MARS		DGAPN	
δ	Improvement	Similarity	Improvement	Similarity	Improvement	Similarity
0	2.24 ± 0.94	0.22 ± 0.06	0.98 ± 1.50	0.15 ± 0.05	$\mathbf{2.79} \pm 1.60$	$\mathbf{0.27} \pm 0.18$
0.2	1.95 ± 0.92	0.25 ± 0.06	0.25 ± 1.55	0.16 ± 0.07	$\mathbf{2.06} \pm 1.23$	$\mathbf{0.32} \pm 0.16$
0.4	1.04 ± 0.90	0.40 ± 0.09	0.08 ± 1.42	0.17 ± 0.09	$\mathbf{1.17} \pm 0.82$	$\mathbf{0.43} \pm 0.25$
0.6	0.55 ± 0.78	0.61 ± 0.11	0.06 ± 1.53	0.19 ± 0.09	$\mathbf{0.58} \pm 0.60$	$\mathbf{0.67} \pm 0.24$

Multi-objective optimization

The docking reward is weighted with a formula including QED [3] and SA [4].

$$r_{m'}(s_T) = \omega \cdot r_m(s_T) + (1-\omega) \cdot \mu_1 \cdot \left[QED(s_T) + \left(1 - \frac{SA\{s_T\} - 1}{\mu_2}\right) \right]$$

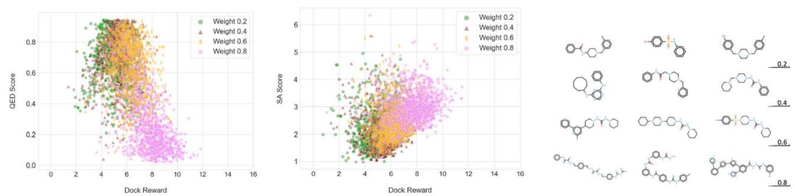


Figure 4: Summary of the molecules obtained under different settings of weight ω. Left and middle plots are QED and SA vs. dock rewards of each 1,000 molecules generated by DGAPN. Right plot shows the top 3 molecules with the highest objective scores generated under each setting of ω.

The balancing between the main objective and realism metrics is investigated to generate promising chemicals in practice with room for hit optimization

Conclusions

Our framework advances the state-of-the-art algorithms in the optimization of molecules with antiviral potential while maintaining reasonable synthetic accessibility. A valuable extension of our work would be to focus on lead-optimization — the refinement of molecular fragments already known to bind the protein of interest through position-constrained modification.

References

Burda, Yuri, et al. "Exploration by random network distillation." arXiv preprint arXiv:1810.12894 (2018).
Schulman, John, et al. "Proximal policy optimization algorithms." arXiv preprint arXiv:1707.06347 (2017).
Bickerton, G. Richard, et al. "Quantifying the chemical beauty of drugs." Nature chemistry 4.2 (2012): 90-98.
Ertl, Peter, and Ansgar Schuffenhauer. "Estimation of synthetic accessibility score of drug-like molecules based on molecular complexity and fragment contributions." Journal of cheminformatics 1.1 (2009): 1-11.

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